

Florian Frick

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EDUCATION

Master of Science in Computer Science

Aug 2023 — May 2025

University of Colorado, Boulder — GPA 3.94

Bachelor of Science in Computer Science

Aug 2020 — May 2024

University of Colorado, Boulder — GPA 3.87

Certificate of Engineering Leadership

EXPERIENCE

Trimble Inc. - Senior Capstone Project

Westminster, CO

Student Consultant

Aug 2023 — May 2024

- Led a team of 5 in creating an automatic calibration procedure for the 3D radar system of Trimble Inc.'s autonomous construction vehicles to reduce calibration time and eliminate the need for an on-site expert.
- Used Python and ROS2 to implement DBSCAN clustering and Procrustes superimposition to extract features and perform point alignment on noisy data to achieve a transformation error of down to 0.1m between radars.
- Followed Agile methodology, using Jira, Docker, and Github for CI/CD and collaborative development.

PROJECTS

NBA Sentiment Dashboard

2026

- Engineered a full-stack analytics platform using React, DynamoDB, and AWS AppSync to visualize trends and correlations between player statistics and fan sentiment.
- Built an automatic pipeline to scrape Reddit threads, perform one-shot sentiment analysis using Gemini models, and make API requests for real-time updates of player performance.
- Created an agentic RAG AI chat feature using WebSockets and AWS Lambda that can answer complex user questions by combining real-time data retrieval with generative AI capabilities. Implemented conversation history and thought process streaming for improved user experience and transparency.

Imitation Learning for Robot Action Planning

2025

- Used PyTorch to implement and train an encoder-decoder transformer model to autonomously manipulate a T-shaped block with precision with object detection and action planning.
- Matched the performance of the state of the art Diffusion Policy model of 78% task completion while using 30% fewer parameters.

Semantic Segmentation of Arctic Sea Ice

2024

- Trained a Hugging Face transformer model to automatically generate accurate sea ice concentration maps from satellite images of the Hudson Bay with Python using computer vision.
- Improved performance over existing U-net models from 83% to 89% pixel accuracy and 0.44 to 0.50 IoU.

Analysis of Premier League Betting Odds

2024

- Created random forest and logistic regression models to analyze the predictive power of bookmakers' betting odds and player stats for EPL match outcomes in Python with 57% and 53% model accuracies, respectively.
- Identified "winning" strategies, and team/player trends and outliers.

Comparison of Vaccination Strategies on Contact Network

2023

- Compared the performance of vaccination strategies using SIRV simulations on a real contact network of university students to inform public health policy.
- Discovered that targeting high-degree nodes lowers peak and total infections by up to 21% and 50%, respectively, over a strategy that targets bridge nodes.

SKILLS

Languages: Python, Java, Javascript, SQL, C++

Technologies: Linux, Windows, Docker, Git, ROS2, CUDA, PyTorch, React, Flask, NodeJS, AWS

Certificates: AWS Cloud Practitioner (November 2025)